UNIVERSITY OF ENGINEERING & TECHNOLOGY MARDAN

COMPUTER SCIENCE

SUBJECT: COMPUTER NETWORKS

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SECTION : A

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LAB : 2

Computer Networking Lab Installing and Using Wireshark

Introduction :

Wireshark is used for network packet analysis. It helps capture and inspect data packets traveling across a network.

Wiresharks Allows US to:

Troubleshoot network issues: Identify slow or problematic network behavior.

Analyze network security: Detect suspicious traffic or potential security breaches.

Learn about network protocols: Understand how data is structured and transmitted between devices.

Optimize network performance: Spot bottlenecks and inefficient data transfers.

Debug applications: Track how data is being sent and received by software programs.

It's widely used by network administrators, security analysts, and developers.

I Downloaded the **Wireshark** installer from https://www.**wireshark**.org/download.html and execute it.

Icon Of Wireshark

TASK 1

Recording network-related information about your own computer

- Open command prompt
- Run the following command

```
ipconfig /all
```

- Fill out the following table.

S. No	Node/Device	IP Address
1	Your computer	192.168.188.16
2	DNS Server	127.0.2.2
3	Default Gateway	127.0.2.2

Screenshot of the out of the command `ipconfig /all` and paste it below.

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Dual Band Wireless-AC 8260
   Physical Address. . . . . . . . . : E4-A7-A0-95-89-C7
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::6e0a:2ae5:46b9:a0c9%20(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.188.16(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Wednesday, February 12, 2025 12:14:25 PM
   Lease Expires . . . . . . . . . . : Wednesday, February 12, 2025 1:31:14 PM
   Default Gateway . . . . . . . . . : 192.168.188.204
   DHCP Server . . . . . . . . . . . : 192.168.188.204
   DHCPv6 IAID . . . . . . . . . . . : 350529440
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2E-BD-2E-CD-34-4B-50-00-00-00
   DNS Servers . . . . . . . . . . . : ::ffff:127.0.2.2
                                       ::ffff:127.0.2.3
                                       127.0.2.2
                                       127.0.2.3
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

TASK 2

Using Wireshark

- Launch Wireshark and start capturing traffic.
- Open various websites.
- Stop the capture and fill out the following tables.

Fill out information about IP addresses in the following table.

S. No	IP Address	Write whether it appears under source or destination	In indicate whether this IP address is for DNS Server, Default Gateway or some website.
1	192.168.188.16	source	IP Adress
2	52.168.117.171	source	Other
3	192.168.188.101	Destination	Other
4	131.253.33.254	source	Other
5	192.168.188.16	Destination	IP Adress

TASK 3

QUESTION

Fill out information about the protocols that appeared in your capture. Feel free to use google search for finding out the layer to which a protocol belongs.

ANSWER

S. No	Protocol Name	Write the layer name to which this protocol belongs
1	TCP	Transport Layer
2	HTTP	Application layer
3	TLSv1.2	Transport Layer
4	ARP	Data Link Layer

THE END